Franci Rrapi

AI and Robotics Engineer

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PROFESSIONAL EXPERIENCE

PhD Candidate | University of Udine

Udine, Italy • Nov 2025 – Present

- AI applied to human-robot collaboration (HRC) for industrial environments.
- Evaluated multiple Large Language Models (LLMs) on collaborative assembly tasks.
- Secondment of 3 months at Chiang Mai University (within SME5.0 european project) focusing on advanced robotic grasping from 3D Point Cloud captured with depth camera.
- Robot-assisted assembly leveraging zero-shot capabilities of Vision Language Models (VLMs) for task planning and visual reasoning.

Research Fellow | AILAB

Udine, Italy • Jul 2024 – Jun 2025

- Research project 'Advanced artificial intelligence and visual reasoning approaches for collaborative robotics (AI4CR)'.
- Extensive literature review on AI and LLMs for collaborative robotics.
- Experimental tests with a UR5e cobot equipped with an RG6 gripper and a RealSense RGB-D camera for robot-assisted collaborative tasks.
- Developed a framework that applies HRC methodologies incorporating depth cameras for computer vision, LLM/VLM reasoning for task planning on robot-assisted assembly.

QA Engineer | Concept Reply

Italy • Nov 2022 – Aug 2023

- Implemented automated testing frameworks over different test cases, performed performance evaluations, and applied innovative AI and Computer Vision solutions to ensure product quality.
- Supervised thesis students on R&D projects involving a robotic arm, driving successful project completions and laboratory research advancements.

Robotics Engineer (Erasmus+ Traineeship) | Procter & Gamble

Brussels, Belgium • Mar 2022 – Jul 2022

- Research thesis during Msc degree within the AIPlan4EU european project, collaborating with Procter & Gamble Services Company and Sapienza University of Rome.
- Experimental tests on simulated (Gazebo and RViz) and real lab environment with a UR5e robotic arm equipped with a Robotiq 2F-85 gripper and a RGB camera mounted on the gripper.
- Object detection and class recognition with YOLO, enabling automated quality measurement experiments with an integrated AI Planning framework.
- YOLO fine-tuning for a better accuracy in clutter and darkened lab environments.

NLP Engineer (Internship) | Babelscape

Rome, Italy • Sep 2019 – Nov 2019

- Built a data processing pipeline to understand and disambiguate the sentence words meaning.
- Used Neural Networks (e.g., CNN, LSTM) and AI tools to detect and analyze texts written in human natural language (e.g., NLTK Word2Vec, SpaCy).

EDUCATION

MSc in Artificial Intelligence and Robotics | Sapienza University of Rome

Rome, Italy • Sep 2018 – Jul 2022

- Erasmus+ Traineeship | Procter & Gamble | Brussels, Belgium | Mar 2022 – Jul 2022
- Extra-EU Mobility Exchange | UFMG | Belo Horizonte, Brazil | May 2021 – Sep 2021

BSc in Computer Science | Sapienza University of Rome

Rome, Italy • Sep 2014 – Oct 2018

- Erasmus+ Exchange Program | University of Málaga (UMA) | Málaga, Spain | Sep 2017 – Jul 2018

SKILLS & TECHNICAL PROFICIENCIES

- **Programming Languages:** Python, C/C++, MATLAB, Java, JavaScript, SCALA
- **AI & Machine Learning:** TensorFlow, PyTorch, Keras, Vision Language Models (VLMs), SciPy, Scikit-learn
- **Computer Vision:** OpenCV, Tesseract, OCR, YOLO, OWL-ViT, SAM3
- **Robotics:** ROS, Gazebo, RViz, V-REP, MoveIt!, RTDE, SLAM
- **Data Analysis:** MySQL, NumPy, Matplotlib, Pandas
- **Developer Tools & OS:** Git, LaTeX, Colab, Jupyter, Linux, Windows, Docker, Selenium, Cucumber
- **Spoken Languages:** Italian (Native/Bilingual), Albanian (Native), English (Fluent), Portuguese, Spanish, French

PUBLICATIONS

- **Rrapi, F.**, Portelli, B., Serra, G., & Scalera, L. (2026). From AI foundations to Large Language Models: A survey on challenges and opportunities in collaborative robotics. Robotics and Computer-Integrated Manufacturing, 100, 103269.
- **Rrapi, F.**, Portelli, B., Serra, G., & Scalera, L. (2026, June). An Experimental Comparison of Visual Language Models in Robotic-Assisted Assembly. In IFToMM Symposium on Mechanism Design for Robotics (pp. 351-359). Springer Nature Switzerland.
- Iocchi, L., Lomuscio, A., **Rrapi, F.**, Zonfrilli, F., et al. (2022, November). Modelling automated industrial experiments as incremental diverse AI planning problems. AIRO-AixIA workshop.
- **Rrapi, F.**, & Iocchi, L. (2022, July). Robot Control and Plan Execution for Automated Experiments in Industrial Laboratory. MSc Thesis, Sapienza University of Rome.
- **Rrapi, F**., & Nardi, D. (2018, October). Analysis of Sound and Spoken Language for Human-Machine Interaction. BSc Thesis, Sapienza University of Rome.

EXTRACURRICULAR ACTIVITIES

- OW Diver | PADI Certification, Thailand (Jun 2026 – PRESENT)
- Soccer Player | Amatori Ancona 2, Udine (2025 - PRESENT)
- Farmer Volunteering | Fazenda Boa Esperança, Delfim Moreira, Brazil (Jan 2024 – Feb 2024)
- Sales Promoter | MediaWorld, Rome, Italy (Nov 2019 – Apr 2021)
- Library Assistant | DIAG Sapienza, Rome, Italy (Sep 2016 – Aug 2018)

I authorize the processing of my personal data contained in my curriculum vitae according to Legislative Decree 196/2003 and Regulation (EU) 2016/679 (GDPR).